

Rotor speeds from ego-noise: the wrap-up

Figures and tables on one frozen protocol

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The question and the paper

Can we get per-rotor speeds from onboard audio alone?

- Per-rotor speeds from onboard audio: each rotor at rate f leaves harmonics at $f, 2f, 3f, \dots$
- No published direct method exists — we build the baseline suite ourselves.
- A 26-variant architecture search gives three winner architectures.
- Augmentations and a full-envelope real training regime repair what models actually read.
- A blind two-stage classical tracker sets the cruise precision ceiling, at high CPU cost.
- Synthetic data supplies coverage, not realism — and loses its value once real data supplies the same coverage.

Data split

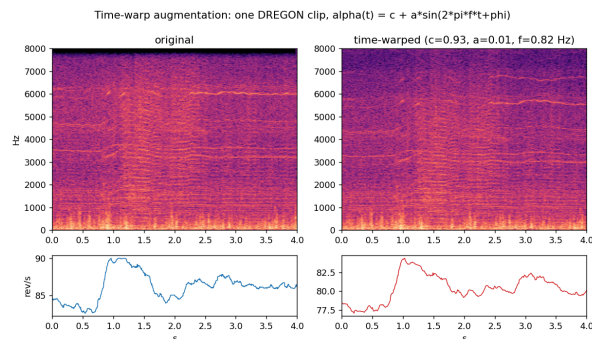
Split	DREGON (small quad)	MD2 (DJI M100)	Speech
Training	free-flight and hover recordings, refined per-rotor labels	FLY124, FLY125 (recalibrated telemetry)	LibriSpeech
Validation (frozen)	8 held-out slices, full envelope (ground, ramps, cruise)	29 held-out slices from FLY124/FLY125	—
Test (reserved, untouched)	free-flight speech-high room1, whitenoise-high room1	FLY103, FLY108 (calibrated, frames published)	—

The validation set: 37 clips $\times$ 8 channels $\times$ 8 s. Per-frame Hungarian matching of 4 predicted to 4 true speeds; errors pooled per regime — **zero** (all rotors stopped), **ramps**, **flight** (all rotors ≥ 45 rev/s) — and overall (frame-weighted).

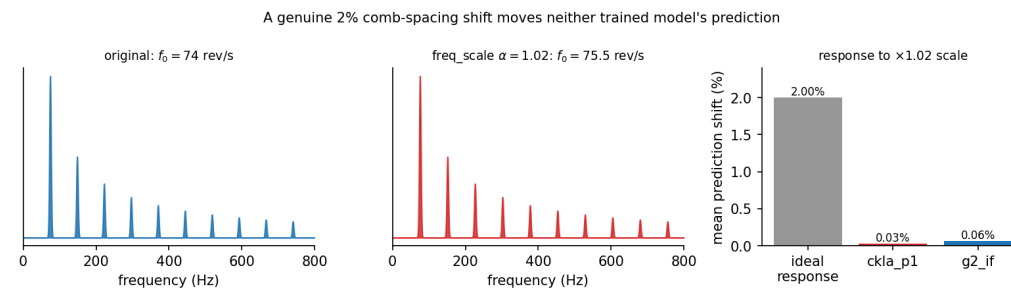
Training regimes

Regime	Full description
R1 — architecture search	Online mixing: random 1-s noise chunks from the real pool, random speech overlay at random SNR, new mixtures every epoch. Augmentations after a 50k-sample warm-up: random gain, polarity flip, channel drop (each $p = 0.5$). No silence, no label-transforming augmentations.
R2 — full-envelope real (final real-only)	R1 mixing plus: silence arm — 16.7% of chunks are synthetic room tone / colored noise / low-frequency rumble with all-zero speed labels; SNR reference floor — quiet chunks are not amplified to the target SNR (reference RMS floor 0.02), so level stops predicting speed; label-transforming frequency-scale and time-warp augmentations (labels scaled/warped together with audio).
R3 — generated + comb curriculum	Stage 1: pre-train on synthetic noise from the neural generator plus the analytic comb, full-envelope speed trajectories with exact silence. Stage 2: fine-tune under R2. Same optimizer, loss, budget.
R4 — comb-only curriculum	R3 with the neural generator removed: stage 1 is the analytic comb alone, stage 2 is R2.
R5 — mixed, one stage	One pool, one stage: real 1/2, generated 1/4, comb 1/4, plus the R2 silence arm; the same augmentation schedule. No curriculum.

Label-transforming augmentations



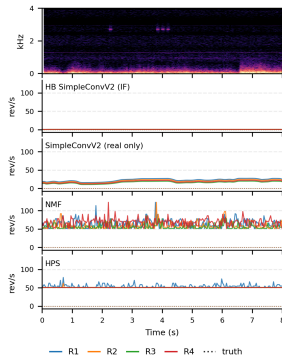
Time-warp: resample audio and speed labels on one warped clock ($\alpha \leq 1.12$) — new speed trajectories from the same recording.



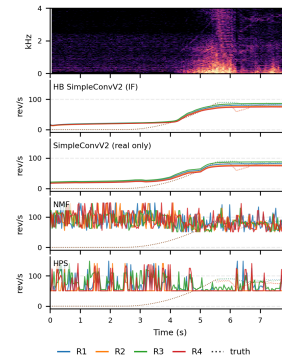
Frequency-scale: scale the spectrum and the labels by the same α — a genuine comb-spacing change the model must follow.

The three regimes of the validation set

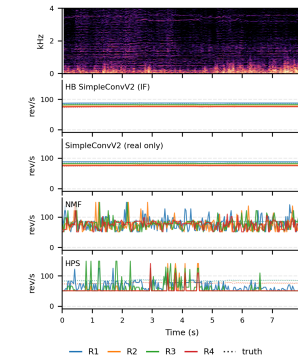
zero — stopped rotors



ramps — start/stop transition



flight — cruise



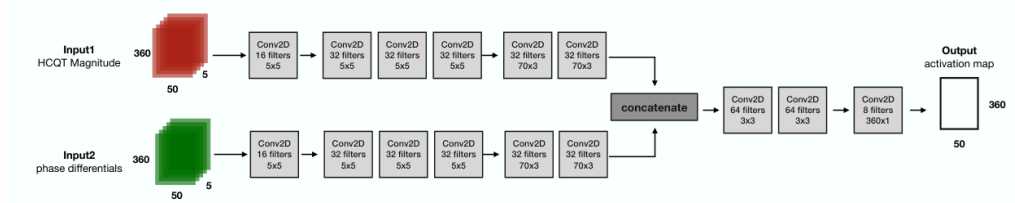
Validation slices: spectrogram on top, per-rotor speed tracks below. The same three clips return full-size later in the deck.

Classical pitch-tracking baselines

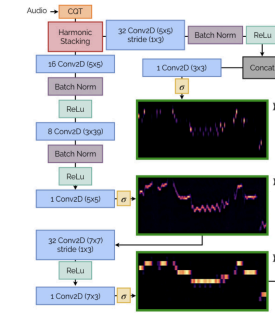
Method	What it is	zero	ramps	cruise	all
PYIN $\times 4$	probabilistic YIN, greedy 4-peak pick	90.8	46.7	34.1	43.0
Cepstral	cepstral peak picking, greedy	95.5	67.5	19.6	35.7
HPS	harmonic product spectrum, greedy	64.1	27.6	20.9	27.3
Matched filter	comb-template bank correlation	87.2	66.8	30.4	42.5
NMF	non-negative factorization, harmonic dictionary	83.8	59.5	8.1	24.6
IHC	inverse harmonic clustering (Björkman & Elvander)	68.8	28.3	16.3	24.6

PIT MAE, rev/s. NMF is the best training-free cruise tracker (8.1). Every method floors at its grid boundary on stopped rotors — silence is unrepresentable for a pitch grid that starts at 50 rev/s.

Salience baselines: what they are

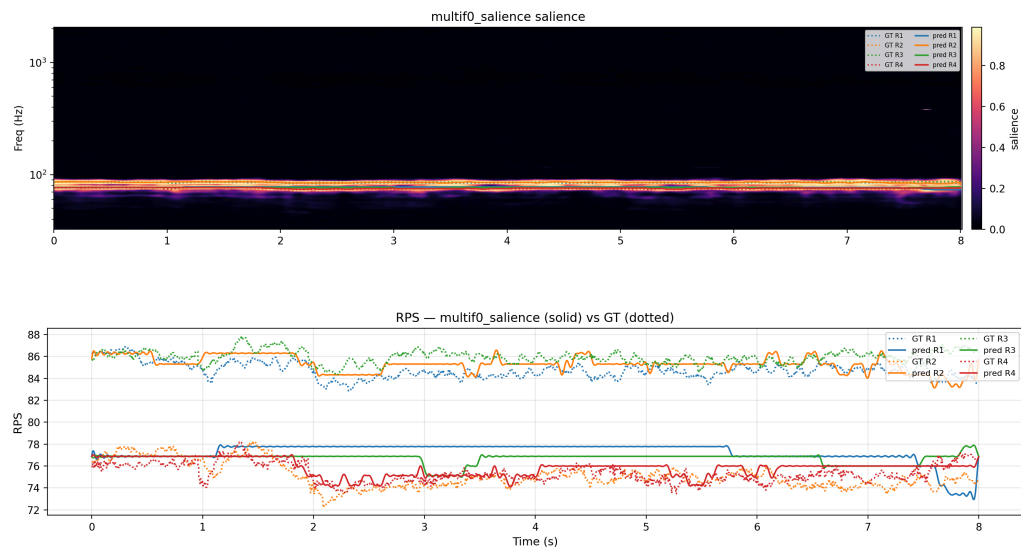


Multi-F0 CNN (Cuesta et al.): HCQT input, per-bin salience map, trained with BCE on the comb positions; Hungarian tracking decodes 4 speed tracks from the map.



Basic Pitch (Spotify): lightweight polyphonic pitch model, ported to PyTorch; same salience-map decoding. Both retrained under the same R2 stream and augmentations as our models.

Salience baselines: outputs and results

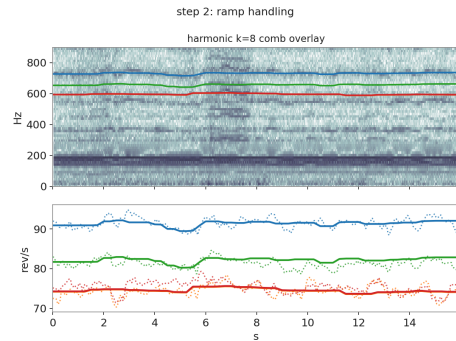


Multi-F0 salience map (top) and decoded tracks (bottom) on a cruise clip.

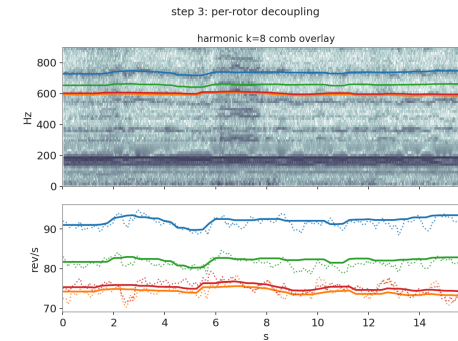
Retrained model	zero	ramps	cruise	all
Basic Pitch	34.0	13.3	31.7	29.5
multi-F0, standard grid	52.8	21.0	4.0	12.5
multi-F0, wide fine grid	48.2	16.1	4.7	11.7

Cruise-capable (4.0 rev/s) and silence-blind: the salience map lights up on room rumble, and no retraining regime fixes it — a model-class limitation, not a data one.

Our classical tracking: blind two-stage annotation



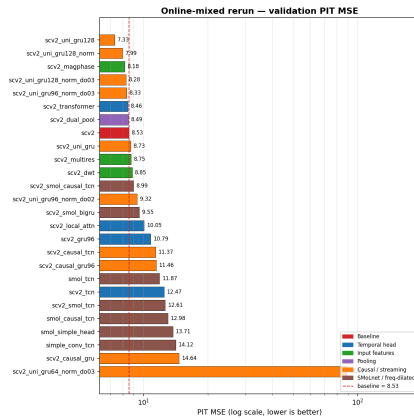
Stage 1 — Viterbi ridge-seeking: comb-score lattice over candidate speeds; joint 8-channel dynamic programming picks the smoothest high-score path per rotor.



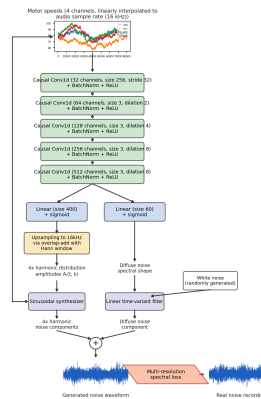
Stage 2 — phase-increment Kalman refinement: demodulate each harmonic at the current estimate, read the phase increments, fuse them with SNR-dependent weights, smooth, repeat coarse-to-fine.

Zero gate: a window is accepted only when the comb score clears its off-comb floor (margin and clearance gates); a refused window emits zero speeds. Gate-free: cruise MAE **2.27** rev/s (best of all methods). Gated: zero MAE **0.01**, at the price of refusing hard flight windows.

Architecture search: 26 variants, 52 runs

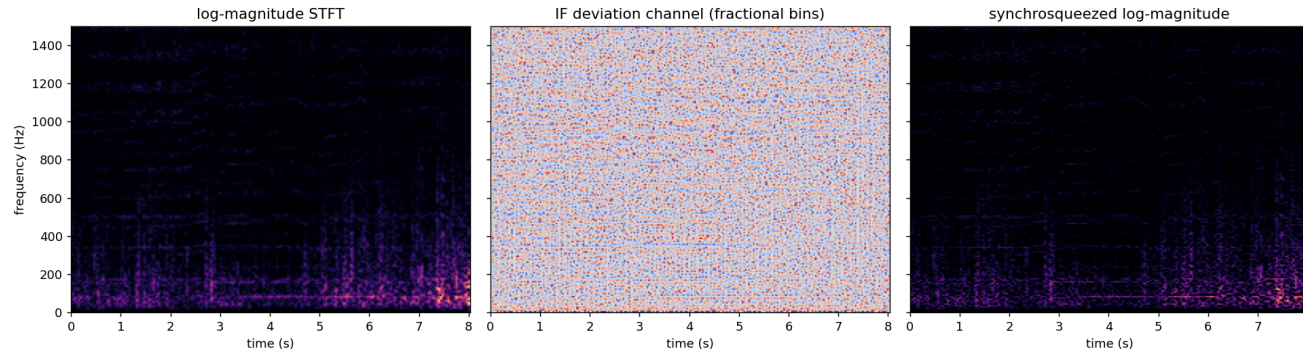


The R1 sweep leaderboard (online-mixing arm): six variant groups over one conv trunk — temporal heads, input features, pooling, causal heads, dilated backbones.



The shared trunk: residual SE conv encoder + frequency-attention pool + temporal head. The three winners kept for the paper: **SimpleConvV2** (BiGRU head), **Transformer head**, **causal GRU-128** (streaming-capable).

Spectral front-ends



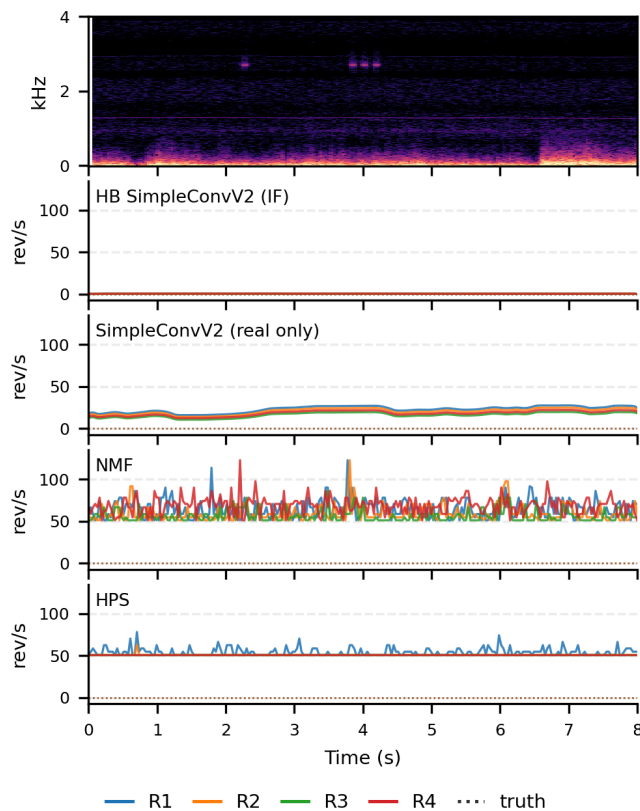
One cruise validation clip through the three front-ends. **Log-magnitude**: the default; comb teeth blurred by window leakage. **IF deviation**: per-bin instantaneous-frequency offset in fractional bins phase information the magnitude discards; zero-mean on noise, structured on tones. **Synchrosqueezed magnitude**: bin power reassigned to the frequency where its phase actually advances — leakage collapses back into sharp ridges, one channel. Best per architecture: IF for SimpleConvV2, magnitude for the Transformer, synchrosqueezed for the causal GRU.

The leaderboard: every family on one protocol

Method	zero	ramps	cruise	all	cost /audio-s
NMF (best training-free)	83.8	59.5	8.1	24.6	0.1 CPU-s
multi-F0 wide grid (best salience)	48.2	16.1	4.7	11.7	2 CPU-s
blind tracker, no gates	79.4	39.1	2.27	17.0	9.9 CPU-s
blind tracker, gates, refusal $\rightarrow 0$	0.01	29.8	48.4	39.7	9.9 CPU-s
SimpleConvV2 (BiGRU), R2	3.4	4.2	2.35	2.72	0.25 CPU-s
Transformer head, R2	5.5	5.1	2.65	3.35	0.25 CPU-s
causal GRU-128, R2	6.0	8.2	3.06	4.12	0.25 CPU-s

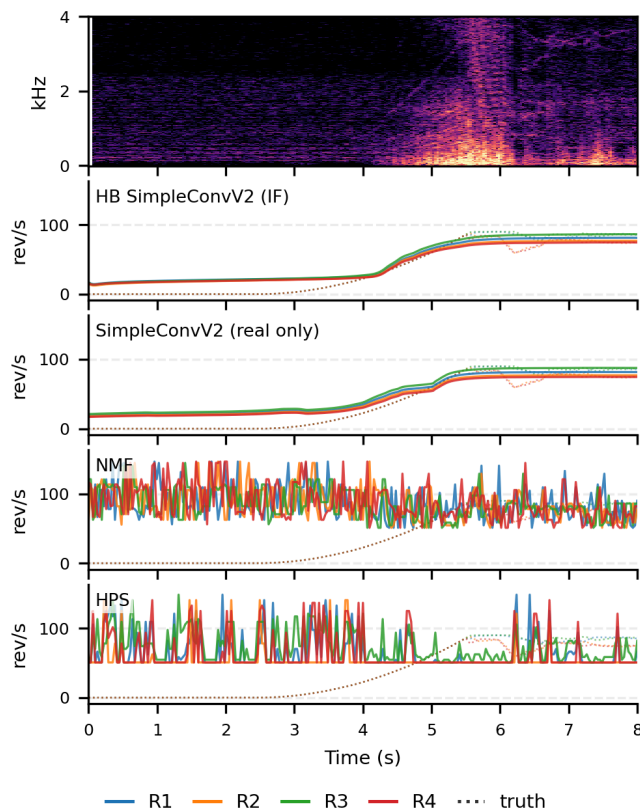
PIT MAE, rev/s. The regression models win every column except cruise, where the gate-free blind tracker stays ahead (2.27 vs 2.35) at $\sim 40\times$ the cost. Only the gated tracker and the R2 models decide silence correctly.

Outputs: stopped rotors



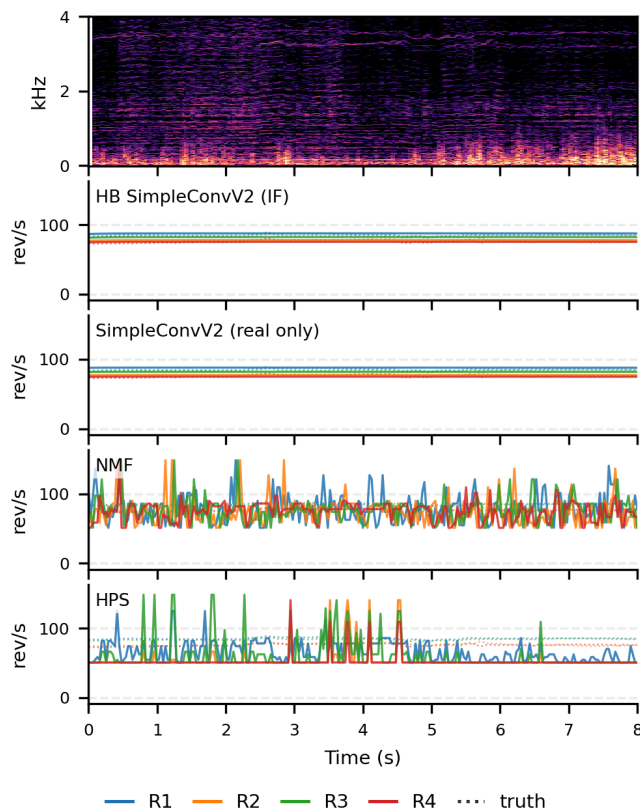
Rumble-heavy silence clip. The R2 model holds zero; NMF and HPS floor at their grid boundary; the cruise-trained model hallucinates speeds from the rumble level.

Outputs: start/stop transition



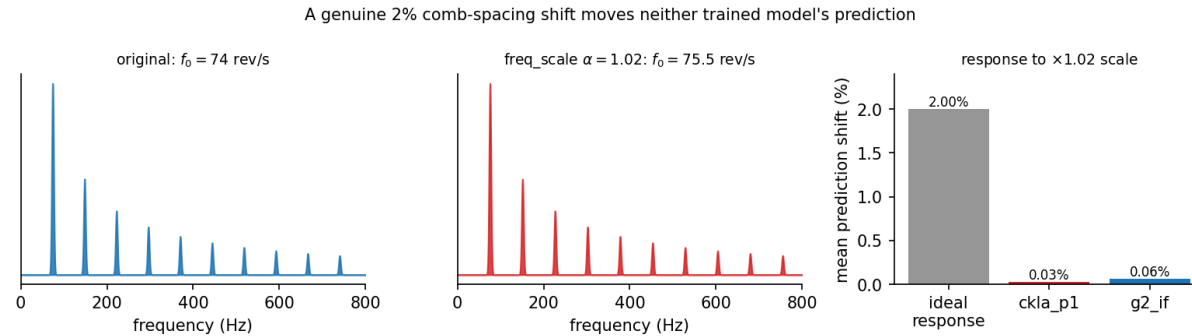
Ramp clip. The full-envelope regime tracks the ramp through the comb collapse; the classical trackers lose lock below their grid floor and re-acquire late.

Outputs: cruise



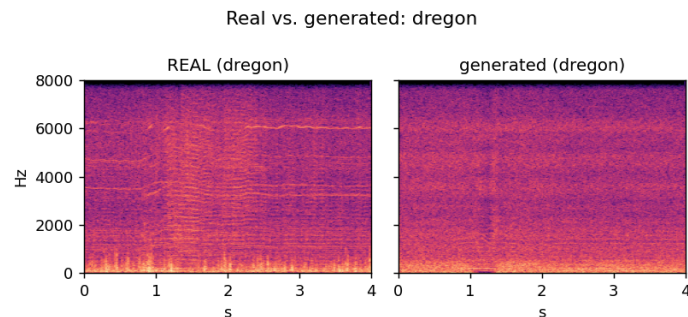
Cruise clip. All methods track; the differences are precision: blind tracker 2.27, best neural 2.35, NMF 8.1 rev/s cruise MAE on the full split.

Why the frequency-scale augmentation is mandatory

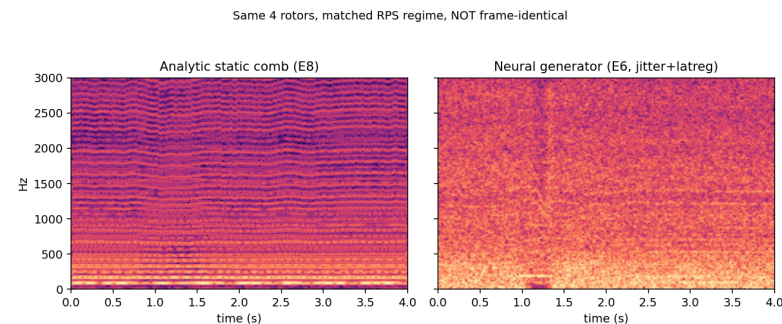


Models trained without it do not read frequency: a genuine 2% comb-spacing shift moves their prediction by 0.03–0.06% — they answer from loudness and the speed prior instead of the comb. The label-transforming frequency-scale augmentation destroys that shortcut and forces the model onto the harmonic structure.

Synthetic data, measured: generator vs comb



Neural generator output vs real DREGON noise: the timbre gap is audible and visible.



The analytic comb: guaranteed harmonic structure at exact labels — no generator training, no timbre modeling.

Pre-training on synthetic noise (R3), then fine-tuning on real data, was the best recipe of the July campaigns. The question the wrap-up answers: does it survive an equally rich **real** regime?

Synthetic data vs the full-envelope real regime

Run (SimpleConvV2 trunk)	zero	ramps	cruise	all
R2 real-only	3.4	4.2	2.35	2.72
R3 generated+comb curriculum	3.9	4.3	2.49	2.90
R5 mixed one-stage	16.1	9.1	5.20	7.11

Aggregate (val MSE)	SimpleConvV2	Transformer	causal GRU
R2 real-only control	22.1	41.8	61.9
R3 curriculum on R2	22.6	—	41.8
R4 comb-only on R2	—	—	37.6
R5 mixed on R2	147.6	59.2	85.8

For the best architecture the curriculum adds **nothing** (22.6 vs 22.1); mixed one-stage training loses on **all three** trunks. Synthetic data survives only for the weakest (causal) trunk, and there the free analytic comb beats the trained neural generator.

The verdict on generated data

- The July gains of the generated-first curriculum came from **coverage**: honest zeros and full-envelope trajectories the real corpus lacked — not from acoustic realism.
- The full-envelope real regime supplies the same coverage from real recordings alone (silence arm + SNR floor + label-transforming augmentations) — and the curriculum's advantage disappears.
- Mixing synthetic data into the real pool acts as label noise and degrades every architecture; staging is mandatory wherever synthetic data is used at all.
- Where a curriculum still pays (causal GRU), the analytic comb — zero training cost — beats the neural generator.